BACKGROUND PAPER

Land for Solar and Wind Development on Oʻahu: Use Rules, Ownership, Grid Proximity, and Terrain

Prepared for Michael J. Roberts (UH Mānoa) · July 12, 2026
Compiled by Claude from primary sources — statutes, ordinances, permit dockets, county tax rolls, and GIS layers. Every figure regenerates from cached data (`analysis/`); full citations are in the repository `notes/` files. Items not confirmed against a primary source are tagged `U`.

Summary. This paper documents the land available for utility-scale solar and wind development on Oʻahu: the use rules that apply to each class of land, who owns it, how far it sits from mapped transmission, and how much of it is flat enough to build on. The main quantitative results: about 42,800 acres of agricultural-district land lie within 1 km of a mapped 46 kV+ line; of that, roughly 6,100–11,100 acres of class D/E land (depending on the slope screen) can host solar today with no acreage cap and no special permit, and a further 3,600 acres of class B/C land are eligible by right under the current 10%-of-parcel/20-acre cap. Where projects exceed the cap, the special-use-permit path has approved 7 of 8 applications since 2014, unanimously, with no intervenors, in a median of about six months. Government owns half the agricultural district; Kamehameha Schools owns a quarter of the private half. Outside the agricultural district, physically suitable urban land exists (~11,900 low-improvement acres) but carries urban prices and competing entitlements; the durable non-ag slice — public land, cheap private land, and quarry/landfill sites — is roughly 5,700 acres (~810–1,130 MW). Wind faces a different constraint: Honolulu's 2025 zoning ordinance sets large-turbine setbacks of at least 1.25 miles from residential-district lots, which leaves 36%

of AG/Country-zoned land geometrically eligible and bars repowering of the existing fleet.

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1 Use rules by land class

Hawaiʻi's Land Use Law places every acre in one of four state districts; the agricultural district covers about 120,800 acres of Oʻahu (roughly 32% of the island; statewide, agricultural districting covers ~47% of Hawaiʻi's land). Within the district, solar rules key off the Land Study Bureau's soil productivity ratings, A (best) through E. Four layers of law apply to a project: the state district rules (HRS ch. 205), county zoning (the Honolulu Land Use Ordinance), county real-property taxation, and — for any project selling to the grid — a power-purchase agreement with Hawaiian Electric approved by the PUC. Table 1 states the current state-district rules for solar; wind is covered in §9.

Table 1. Current state-district rules for solar on agricultural land (HRS §§205 (20)–(21))

Soil class	Solar as a permitted use	Above the cap
A	Not permitted	The §205-4.5(a)(20) proviso bars solar as a permitte class A, and (a)(21) authorizes SUPs on B/C only — bu permitting practice has not treated this as an absolute least one granted SUP (SP15-406, 2015) included cla with panel areas coded on A soils. Recent applicants soils out of project areas; a district boundary amendm unambiguous route

B / C	Up to min(10% of parcel, 20 acres) per parcel	County special use permit (§205-6), with conditions: paneled area must be offered for compatible agricultu ≥50% below fair-market rent; decommissioning secur restoration
D / E	Permitted outright — no acreage cap, no special permit	n/a

Three county-level rules matter alongside the state rules:

- **Zoning.** The Honolulu LUO permits solar in AG districts subject to ordinary development standards; solar faces no county setback regime comparable to wind's (§8).
- **Property tax.** Land under agricultural dedication is assessed at 1–5% of fair-market value. In 2021 the city assessor reclassified land under operating solar farms as industrial (full value, $12.40/$1,000), raising some annual bills about 25-fold mid-contract; a partial exemption for renewable projects (reported as 80%; ordinance text unverified) was subsequently adopted. Assessment treatment remains a material, county-discretionary cost item for project underwriting.
- **Procurement.** All grid sales run through HECO power-purchase agreements awarded in competitive RFP rounds and approved by the PUC. In practice land is optioned and permits filed after an award, so procurement timing governs when land actually enters development.

2 Procedural pathways and timelines

A project's permitting path depends on soil class and size. On D/E soils, and on B/C soils within the cap, solar is a permitted use: county building, grading, and electrical permits apply, and no discretionary land-use approval is required. Above the cap on B/C soils, the special-use-permit path runs through the county planning commission and, for permits over 15 acres, the state Land Use Commission. A complete census of the LUC's special-permit series (1963–2026) contains eight solar dockets, all filed since 2014:

Table 2. Solar special use permits reaching the Land Use Commission, 2014–

census)

Statistic	Value
Dockets	8 (SP15-405/406/407, SP17-408, SP21-411/412, SP26-
Outcomes	7 approved, 1 pending, 0 denied, 0 withdrawn; every LUC
Intervenors	0 of 8; organized testimony in the dockets is supportive (u Ulupono)
Median duration	County stage 141.5 days; LUC stage 34 days; total ≈ 6 mo
Applicants and landowners	AES on 4 of 8; land owned by Kamehameha Schools, Grov McBryde/A&B, Robinson, Mahi Pono, Castle & Cooke, Mon
Counsel	Matsubara Kotake & Tabata on 5 of 8

The SUP conditions are inexpensive in practice. The ag co-use clause requires that the paneled area be *made available* for compatible agriculture at half of market rent; grazing qualifies, and grazing leases substitute for vegetation management the operator would otherwise purchase. Decommissioning security and restoration are standard, modest costs for PV. The observed cost of the SUP path is mainly time (≈6 months of discretionary process) and the uncertainty of a case-by-case approval, which lenders price. To date every applicant has been a major developer on a large landowner's parcel. Smaller landholders on B/C soil have a 20-acre by-right envelope and would need the SUP process to go beyond it; navigating that process is routine for the repeat developers and law firms in the census, and a developer partner would ordinarily carry it for a lessor.

Where the rules bind therefore depends on parcel size and owner capacity together. On parcels under 200 acres the 20-acre term never binds (10% of the parcel is the smaller number); on parcels above 200 acres the 20-acre term binds, but the owners of most such parcels are governments and large private estates for which the SUP process is routine. Neither restriction binds hard unless the SUP is difficult for the particular owner — which points at mid-size private owners of cap-bound parcels, quantified in §3.

3 Ownership

Ownership of the Oʻahu agricultural district is highly concentrated (Figure 1). Attribution covers 99% of acreage, from the city's bulk assessment table with entity resolution across name variants. Acreage totals here are full parcel areas for parcels intersecting the district (parcels can straddle district boundaries), so they exceed strict in-district acreage; shares are computed on the same basis throughout.

- Government owns 50.1%: State of Hawaiʻi 25.8% (including DLNR and ADC), the United States 19.6% (largely military), the city 2.7%, DHHL 2.0%.
- Kamehameha Schools owns 12.8% of the district — 26.1% of the private half. The twenty largest private owners hold 71.2% of private acreage.
- The 20-acre cap term binds only on parcels above 200 acres with B/C soil: 118 parcels on Oʻahu, led by the United States, Kamehameha Schools, Dole, and the State. Changing the cap therefore changes eligibility mostly on large institutional holdings.

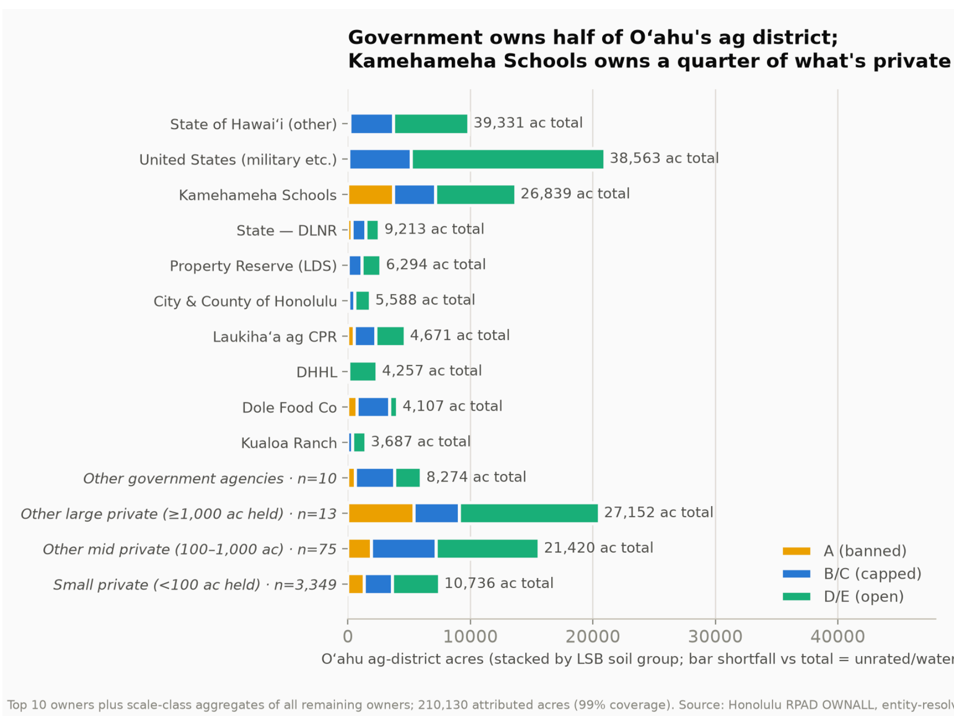

Figure 1. Ownership of the Oʻahu agricultural district: the ten largest owners individually, plus scale-class aggregates covering all remaining owners (other government agencies, and other large / mid / small private owners by total holdings). Stacked by soil group. Source: Honolulu RPAD OWNALL bulk table joined to parcel–soil intersections; 99% of acreage attributed.

Parcel size × owner scale: where the cap binds and for whom the SUP is hard

Table 3 crosses parcel size against owner scale (an owner's total ag-district holdings, with governments broken out) for B/C acreage. Reading it against the rules in §§1–2:

- **Parcels under 20 acres** (10% term binds; ≤2 buildable acres each): B/C acreage here is mostly small-owner land — suited to accessory or CBRE-scale projects, not utility-scale siting.
- **Parcels of 20–200 acres** (10% term binds; 2–20 buildable acres each): about 6,100 B/C acres across all owner classes. By-right projects up to 20 acres are possible at the top of this range.
- **Parcels over 200 acres** (20-acre term binds): 118 parcels, 25,200 B/C acres. Governments hold 49 of these parcels, large and major private owners (≥1,000 ac total holdings) hold 43 — owners for whom the SUP process is documented routine (§2) or for whom commercial leasing runs through developers in any case.
- **The exposed set: 26 parcels, 26 distinct mid-size private owners** (100–1,000 acres total holdings), about 3,700 B/C acres. These owners face the 20-acre cap and are the least likely to carry a discretionary county-plus-LUC process themselves. No small owner (<100 ac held) owns a cap-bound parcel. In practice a developer partner would ordinarily run the SUP for a lessor — every SUP in the census was developer-driven — so the practical question for this group is access to developer interest, which runs through HECO procurement rounds, rather than the permit process itself.

Table 3. B/C acreage by parcel size and owner scale, Oʻahu ag district (acres)

Parcel size	Government	Major private (>5,000 ac held)	Large private (1,000–5,000)	Mid private (100–1,000)
<20 ac	1,033	115	80	231
20–50 ac	514	141	208	74
50–200 ac	2,117	337	704	1,300
200–1,000 ac	4,835	857	1,679	3,674
>1,000 ac	5,107	3,169	5,876	0

4 Grid proximity

Public line data (HIFLD and OpenStreetMap, cross-validated) maps Oʻahu's 138 kV network well (~325 km, both sources agree) and the 46 kV network poorly (~80 km mapped of several hundred km built), so all distances below are upper bounds — true distances are shorter. Line capacity and substation headroom are confidential and are not screened here; to the extent upgrades can ride existing rights-of-way, distance to an existing corridor is the relevant siting variable.

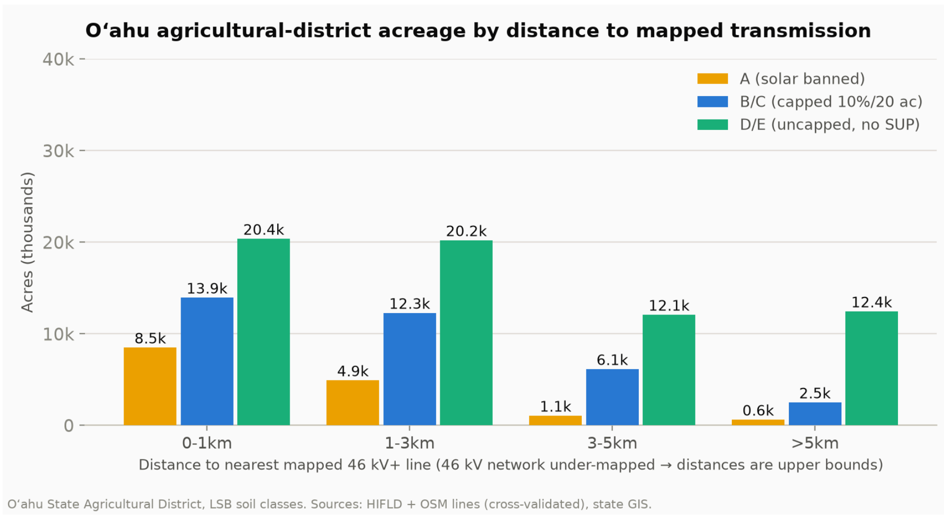

Figure 2. Agricultural-district acreage by distance to the nearest mapped 46 kV+ line and soil group. Within 1 km: 20,359 D/E acres (no cap, no SUP), 13,944 B/C acres (capped), 8,497 class-A acres (solar not permitted). 75% of all B/C acreage lies within 3 km.

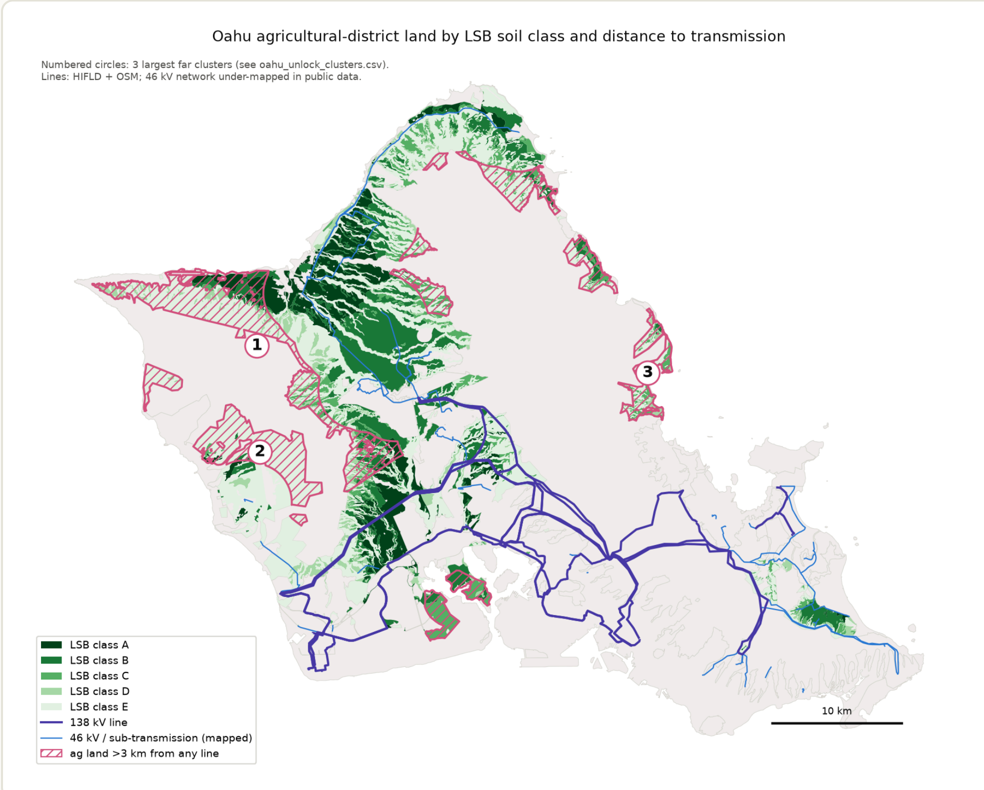

Figure 3. Soil class, mapped lines, and the largest land clusters more than 3 km from any mapped line. The far clusters are Waiʻanae-range hillside (mostly class E), Lualualei valley (largely federal), and Koʻolauloa mauka.

Strategic transmission additions

To locate where new or upgraded lines would add the most access and bidding competition, all buildable (slope-screened) ag-district land more than 1 km from a mapped line was clustered and each cluster scored on two denominators: buildable acres per kilometer of new right-of-way, and distinct landowners served per kilometer (Figure 4, Table 4). Owner count matters for competition: a corridor that reaches five owners' land creates five potential PPA bidders; one that reaches a single estate creates one.

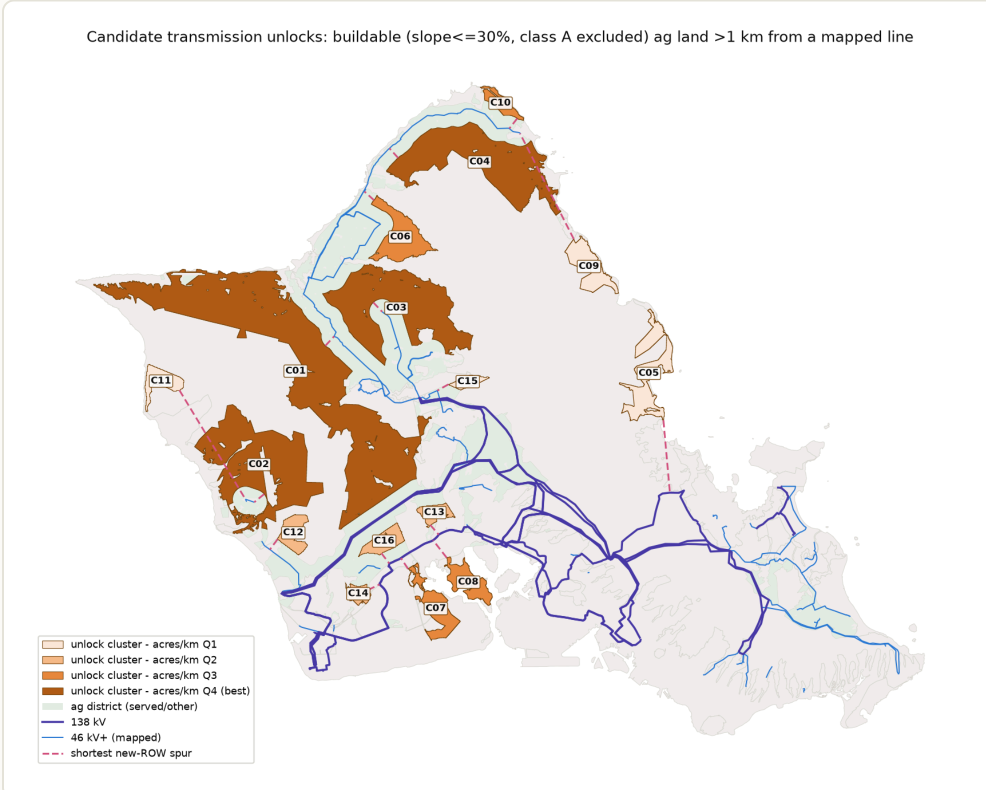

Figure 4. Candidate transmission unlocks: buildable (≤30% slope) ag land beyond 1 km of a mapped line, clustered and shaded by acres-per-km rank, with the shortest new-ROW spur to each cluster (dashed). Class A land is excluded (solar is not permitted on it under any path).

Table 4. Leading corridor candidates and the in-ROW upgrade ring (slope ≤3(MW at 5 ac/MW)

Candidate	Buildable ac	MW	Owners (≥100 ac)	Top owners	Notes
1–3 km ring (upgrade existing ROWs)	≈24,000	≈4,800	28	Dispersed	Served by sł reconductor corridors; laı widest owne option. ≤15‹
C03 Waialua	5,697	1,139	5	Dole, Kamehameha Schools, ag-CPR, State	≈2 km of nev soil, so most contingent c The standou competitive
C04 Kahuku	5,618	1,124	4	US 36%, Property Reserve (LDS) 33%	Partly federa politics appl'

C01 Waiʻanae mauka/plateau	14,210	2,842	18	Mixed	Largest on b military-adja "unserved" s artifact of 46 — verify aga before weigh
C02 Lualualei	8,749	1,750	—	US Navy (majority)	Majority fede leasable only enhanced-u

Two readings follow. Upgrading existing rights-of-way serves more buildable land (≈24,000 non-A acres) across more owners (28) than any greenfield corridor, consistent with treating in-ROW capacity upgrades as the first-order transmission strategy. Among genuinely new corridors, a short (~2 km) Waialua spur is the one candidate that combines volume, multiple private owners, and commercial availability — and its value is mostly contingent on relaxing the B/C cap, since 74% of the cluster is B/C soil. Several apparent clusters (Ewa/Puʻuloa, Waiau, Helemano) are likely artifacts of the 46 kV mapping gap, and the federal-majority clusters are available only through military leasing programs. Line and substation capacity, which is confidential, is unexamined throughout.

How much land a given length of new line can reach

A budget-constrained version of the same question: for a total length L of new line, placed optimally (greedy heuristic, branching from the existing mapped network), how much class B–D land at ≤30% slope comes within 1 km of a line? Figure 5 traces the full curve; Figure 6 maps the first kilometers of the build-out. Class E is excluded here along with class A; routes are geometric, so the curve is an upper bound on what any real routing would achieve at a given length, and the under-mapped 46 kV network means baseline coverage is understated.

- Baseline (existing mapped network): 16,328 of 40,870 eligible acres (40%) are within 1 km.
- The marginal payoff is highest in the first ~11 km (≈400–480 acres per km) and declines steadily after that — diminishing returns are gradual rather than sharp. Ten kilometers of new line raises coverage to 20,185 acres (+3,857); 25 km to 24,195; 50 km to 29,481; 100 km to 36,103 — 88% of everything eligible.

- The first 6 km are three short spurs, all extensions of existing corridors: ~2.2 km mauka from the Waialua 46 kV loop (the same area as corridor candidate C03), ~2.9 km down the Kunia / central-plateau spine south of Schofield, and ~0.6 km at Kahuku — together +2,252 acres. The 25–50 km range thickens Waialua / Kunia and reaches the Waiawa cluster; spending beyond ~75 km buys Mokuleia and remnants.

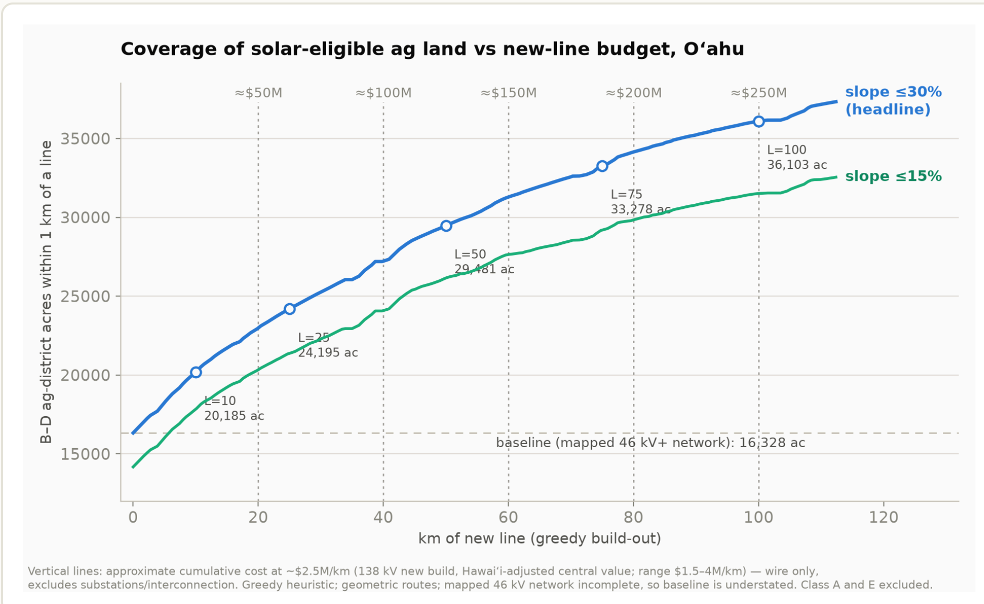

Figure 5. Coverage of B–D ag-district land (≤30% and ≤15% slope) within 1 km of a line, as a function of km of new line placed by the greedy heuristic. Marked points at L = 10/25/50/75/100 km. Vertical lines mark approximate cumulative cost at ~$2.5M/km (138 kV new build, Hawaiʻi-adjusted central value from Table 5; range $1.5–4M/km) — wire only, excluding substation and interconnection costs.

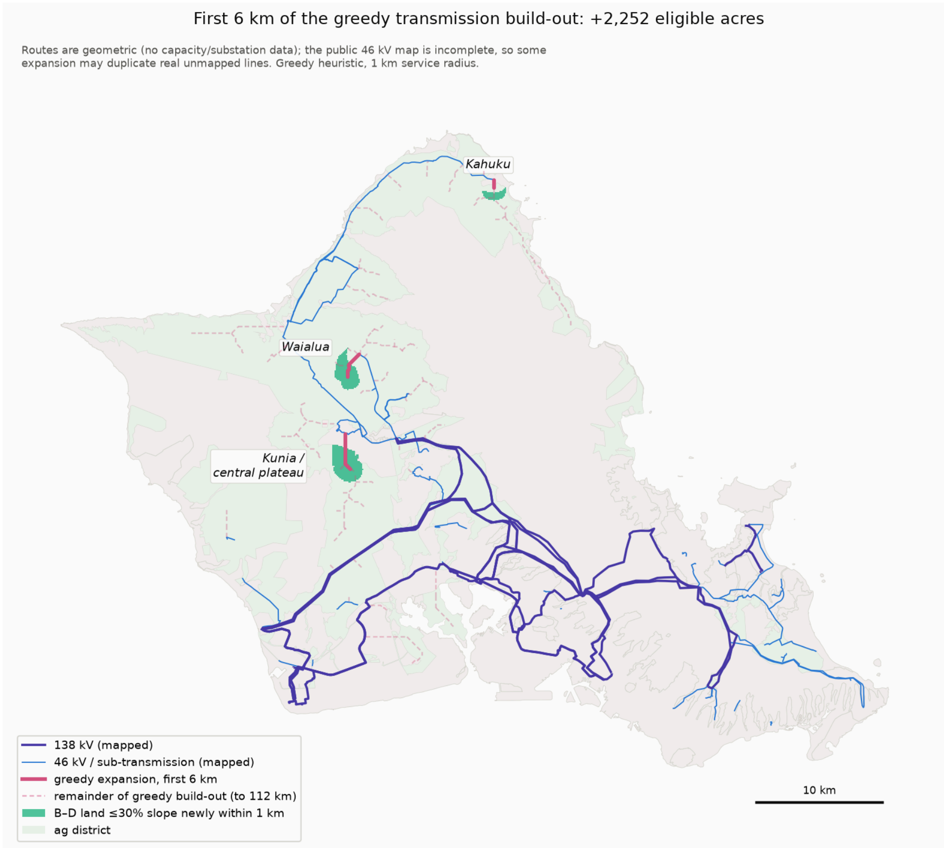

Figure 6. The first ≈6 km of the greedy build-out: three short spurs at Waialua, Kunia, and Kahuku (solid), the remainder of the greedy build-out to 112 km (dashed), and the land newly served (shaded).

Cost benchmarks and an illustrative $500 million program

Planning-level unit costs put the expansion results in dollar terms. Cost per MW-km is not a primitive — it is line cost per km divided by transfer capacity, so it varies an order of magnitude with voltage class. Table 5 gives mainland benchmarks (MISO transmission cost estimation guides, 2024$) with a Hawaiʻi construction multiplier of 1.5–2.5× applied U — HECO's actual unit costs are confidential. Substation work adds fixed costs of several million dollars per connection point, which for spurs this short can exceed the wire itself.

Table 5. Transmission unit-cost benchmarks (mainland planning values per tl Cost Estimation Guide, MTEP24/MTEP25, × 1.5–2.5 Hawaiʻi multiplier; capaci planning-typical values, unverified for HECO equipment)

Class	Build cost, Hawaiʻi-adjusted	Typical transfer capacity	Im
46–69 kV sub-transmission	≈ $0.6–2.5M per km	≈ 20–60 MW	≈ !

138 kV single circuit	≈ $1.5–4M per km	≈ 150–250 MW	≈ $
Substation / collector bay	≈ $10–30M per station	—	fix pc

At these rates, wire length is a second-order project cost: a 3-km spur adds roughly $30–75k/MW to a project whose all-in Hawaiʻi capex is about $2–2.5M/MW — 1.5–3.5%. What a transmission budget mainly buys is *capacity* and *interconnection points*, not route-miles. The greedy curve makes the same point from the land side: about 50 optimally placed km already reaches ~29,500 eligible acres, roughly the ~30,000 acres a fully grid-scale 5–6 GW decarbonization build would need at current densities (later builds at higher MW/acre need less).

Table 6 lays out an illustrative $500 million program built on those observations. Its design principles: rebuild inside existing rights-of-way where possible (new corridors are where organized opposition concentrates); build spurs at 138 kV rather than 46 kV (headroom, not conductor, is the scarce input); and site collector substations by *ownership cluster*, with published available capacity, so that interconnection feasibility becomes visible information and every cluster of independent landowners can produce competing PPA bids.

Table 6. **An illustrative $500M transmission program for Oʻahu solar (plannin from Table 5)**

Tranche	Budget	What it buys	Rationale from this
1. Rebuild the ag-belt sub-transmission at 138 kV in existing ROWs	≈ $150M	50–70 km: Waialua–Wahiawā–Kunia loop + Kahuku segment	The 1–3 km ring hold buildable non-A acre in-ROW work minimi exposure; 138 kV mo of 46 kV for modestl
2. Collector substations with pre-built interconnection bays	≈ $120M	5–6 stations: Waialua, Kunia/central plateau, Kahuku, Waiawa/Waipiʻo, ʻEwa edge	Placed by ownership alone reaches Dole, I State land); publishe converts private inte information into oper opportunities — the c competition in the pr
3. The first spurs, built at 138 kV	≈ $60–80M	20–25 km: Waialua mauka, Kunia spine,	The optimization's hi segments (≈400–48

		Kahuku, then toward Waiawa	
4. Backbone delivery reserve	≈ $120M	Second circuit / rebuild on the central 138 kV path, or state cost-share into HECO's Integrated Grid Planning	Collected power mus Honolulu load center network has no 138 central spine, and HE price full north-to-sc solutions at ≈$1.2–1 is a down payment (s analysis below)
5. Contingency, studies, queue support	≈ $30M	Interconnection studies, hosting-capacity mapping	Study queues are a r priced into bids

Two complements the budget cannot substitute for. First, **cap reform**: most of the acreage this network reaches is B/C soil, SUP-only above 20 acres under current law, so the transmission investment and the one-sentence statutory change are strict complements. Second, **storage co-location at the collectors**: grid-forming batteries shift delivery off-peak and shrink the backbone requirement, so part of tranche 4 is plausibly better spent as storage than wire — a dispatch-modeling question beyond this land analysis, and one for a comprehensive capacity-expansion study to settle. On sufficiency: this program removes land access and bid entry as constraints and makes the remaining constraint — bulk delivery — explicit; at these unit costs $500M is transformative for access and competition and only partial for 5–6 GW of deliverability, which is the right order to spend in.

Bulk delivery to the southern load center

Most Oʻahu demand sits in the southern corridor (Pearl Harbor–downtown–east), and the 138 kV backbone was built to serve it from southwest generation: FERC Form 1 lists 31 circuits over ~200 route-miles at 138 kV — Oʻahu's maximum operating voltage — arranged as two west-anchored corridors from Kahe (one northern via Hālawa–Koʻolau–Pukele, one southern via Waiau and downtown). **There is no 138 kV north of the central spine.** Wahiawā, Waialua, and Kahuku connect only through 46 kV feeders off the Wahiawā and Koʻolau substations, publicly described as at capacity; Wahiawā is a single-bay station not built to export. Published system-level limits bound today's northern deliverability at a couple hundred megawatts: the 2021 renewable-energy-zone study's 135 MW single-point-of-failure limit, and the 2023 IGP finding that full build-out of the northern zone overloads the Wahiawā-area 138 kV system unless injection is cut by 220 MW.

A screenline count of the mapped geometry (Figure 7) makes the same point: two 138 kV circuits and one mapped 46 kV feeder cross the Wahiawā–Waipiʻo band; zero 138 kV circuits cross the North Shore neck. Applying planning-typical thermal ratings (150–250 MVA per 138 kV circuit, 30–60 MVA at 46 kV — guesses; HECO's ratings are confidential) gives the capacity column of Table 7. On the requirement side, on-site storage caps what the corridor must carry: with 4–6 hour batteries sized at 50–100% of plant power, the flow requirement saturates near (southern load share) × (evening load) — roughly 930 MW at today's ~1.2 GW peak, ~1,550 MW under a 2 GW electrified-2045 peak — rather than anything near installed solar capacity.

Table 7. **North→south transfer: guessed existing capability vs requirement (d scenario grid: notes/oahu-bulk-delivery.md, data/gis/screenline_requiremen**

Screenline	Mapped crossings	Guessed capability	Required at GW norther solar
Central plateau → Pearl Harbor (SL-A)	8 geometric 138 kV crossings in 5 corridors (the Kahe circuits loop through the Kunia bench)	≈1,600 MVA central reading (300 conservative)	roughly adequate
Wahiawā → Waipiʻo (SL-B)	2 × 138 kV + 1 mapped 46 kV	≈ 330–560 MVA	500–1,000 → gap up to +800 MW (N
North Shore neck (SL-C)	0 × 138 kV; 2 mapped 46 kV	≈ 60–120 MVA	250–500 M gap +160 tc +470 MW

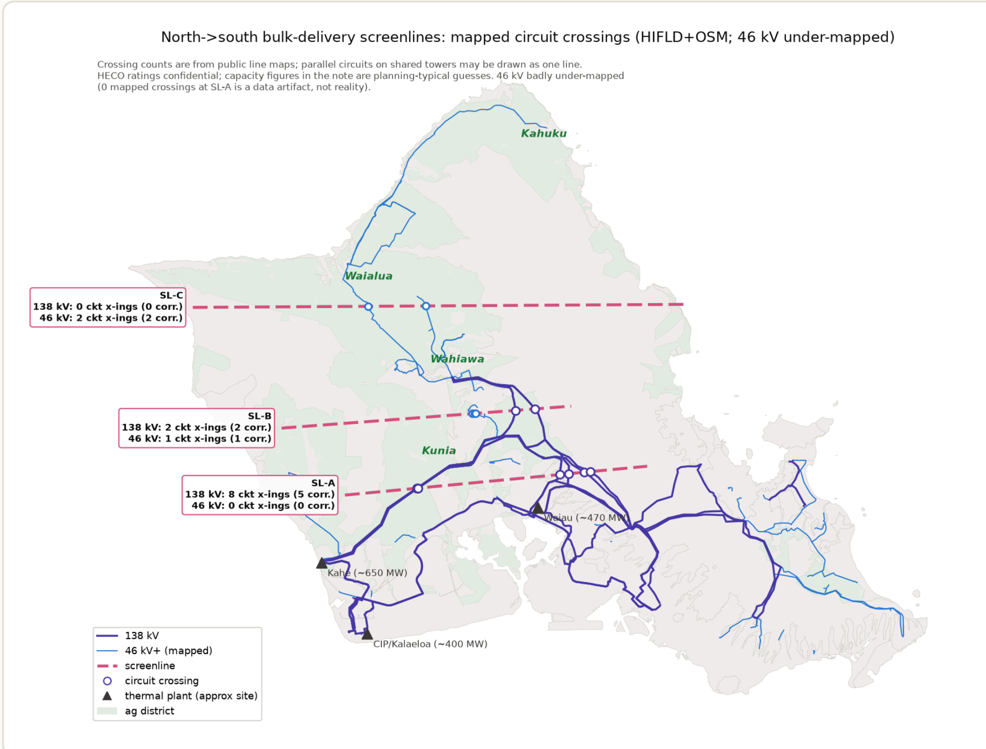

Figure 7. Screenlines and mapped circuit crossings. The backbone runs west→east along the southern corridors; northern land connects over 46 kV only. Ratings are planning-typical guesses; the mapped 46 kV network is incomplete.

Closing the gap does not require a new voltage class. At 10–25 km, thermal limits govern rather than stability, so parallel 138 kV circuits in existing rights-of-way suffice: HTLS reconductoring plus second circuits on the Waialua–Wahiawā, Kahuku–Wahiawā, and Wahiawā–Waiau corridors prices at roughly **$90–250M for a 2 GW northern build and $190–460M cumulative for 4 GW** on this paper's unit costs — about 2–5% of the corresponding solar capex. HECO's own studied options for full north-to-south delivery — new Kahe–Wahiawā 138 kV lines, reconductoring, or a 345 kV Kahe–Wahiawā–Waiau loop — cluster at **≈$1.21–1.28 B**, all within ~5% of each other; the gap between that figure and this paper's geometric estimate reflects scope (substation depth, land, contingency standards, and system-wide rework) that the geometric model does not price, and the near-equality of HECO's three options suggests common fixed costs dominate. Both numbers are reported; neither substitutes for a power-flow study.

Further investigation, in order of value: (1) the HNEI/GridSTART route — HNEI co-owns the validated GE MAPS/PSLF Oʻahu network model and publishes bus-level grid-strength rankings (west strongest, Wahiawā weakest); direct collaboration would replace every guessed rating in Table 7; (2) the IGP docket's transmission analyses (2018-0165) and the 2021 REZ study appendices; (3) HECO's RFP interconnection guidance (spare 138 kV terminations are published: Kahe, ʻEwa Nui, Hoʻohana, CEIP — leeward/central has room, windward is full); (4) curtailment-by- cause data as a revealed-constraint measure; (5) a proper N-1 power-flow on a synthetic or HNEI-supplied network. Sources and vintage caveats: notes/oahu-grid-public-record.md and notes/oahu-bulk-delivery.md.

5 Terrain

A 10-meter USGS elevation model, rasterized on the same grid as the parcel, soil, and line layers, adds slope in five-point bands (Figure 8, Table 8). Slope matters differently for the two legal categories. B/C soil is prime agricultural land and is overwhelmingly flat: 89% of the acreage that a relaxed cap would make eligible near the grid sits at or below 15% slope (97% at ≤30%). D/E soil includes large areas of hillside inside the nominal district boundary: 45% of near-grid D/E acreage is steeper than 30%.

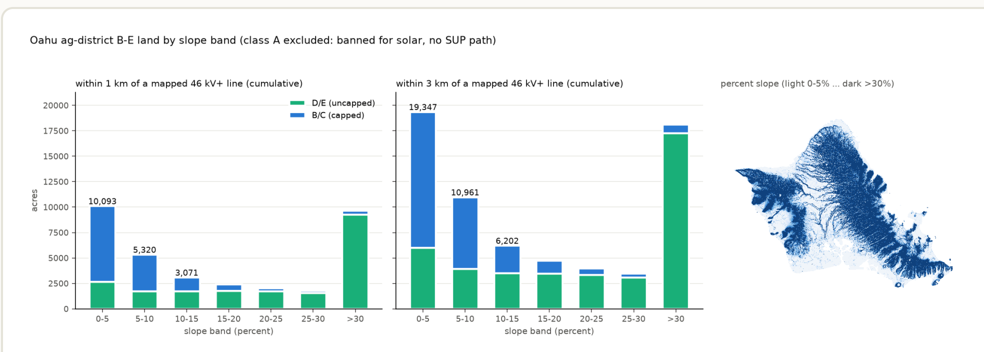

Figure 8. Ag-district land within 1 km and within 3 km of a mapped 46 kV+ line, by percent-slope band and soil group (cumulative panels), with the island-wide slope raster inset. Class A land is excluded.

Table 8. Buildability and cost by slope band (engineering literature), with Oʻa

Slope	Standard tracker	Terrain-following tracker	Indicative cost premium vs flat
0–5%	Yes	Yes	≈ 0 (NREL benchmark reference terrain)
5–10%	Yes	Yes	Low single-digit % of capex derived

10–15%	At spec limit (≈15% N–S)	Yes	Mid single-digit % derived; Honolulu grading code requires an engineer's soils report above 15%
15–20%	No	Yes (built precedent ~20%)	Unpublished; erosion/stormwater costs intensify
20–25%	No	Vendor-reported projects planned	Unpublished; anecdotal only
25–30%	No	Within one vendor's 37% spec; no completed project above 25% found	Unpublished; civil, erosion, and O&M dominated
>30%	No	No	Treat as infeasible

The engineering literature contains no published continuous cost-versus-slope curve: NREL's cost benchmarks assume flat sites and its supply curves apply slope as a binary screen (10% reference, 5% restrictive); mainstream terrain-following trackers top out near 15% grade; feasibility claims above 20% rest on a single vendor's specification. A workable modeling convention: ≤15% as the mainstream envelope, 15–30% as a specialized premium-cost margin, >30% infeasible. Aspect effects at 21°N (north- versus south-facing slopes) remain to be quantified with PVWatts/SAM runs.

6 Combined viability estimates

Putting the screens together for land within 1 km of a mapped line (all figures at 5 acres/MW; 7 acres/MW scales them by 5/7):

- **Available today, no discretionary approvals:** D/E soil at ≤15% slope, 6,083 acres (~1,217 MW); at ≤30% slope, 11,110 acres (~2,222 MW). Plus 3,601 acres of by-right B/C eligibility under the current cap (~720 MW, island-wide; spread in ≤20-acre pieces across many parcels).
- **Available via the SUP path:** B/C acreage above the cap, in practice limited by procurement awards rather than by the permit process (Table 2).
- **Under alternative cap rules** (Figure 9): raising the parcel share to 20% while keeping the 20-acre term adds little (4,743 acres); removing the 20-acre term at 10% yields 9,410 acres; 20% with no acreage term yields

15,657 acres (~3,131 MW), of which 72% is within 1 km of a line and ~90% is at ≤15% slope. The 20-acre term, which binds only on parcels over 200 acres, accounts for most of the difference between scenarios.

Combining cap-reformed B/C with buildable D/E gives a land-side envelope for near-grid utility solar on Oʻahu of roughly 16,000–21,000 acres — order 3–4 GW — before flood, aspect, cultural-resource, and interconnection-capacity screens.

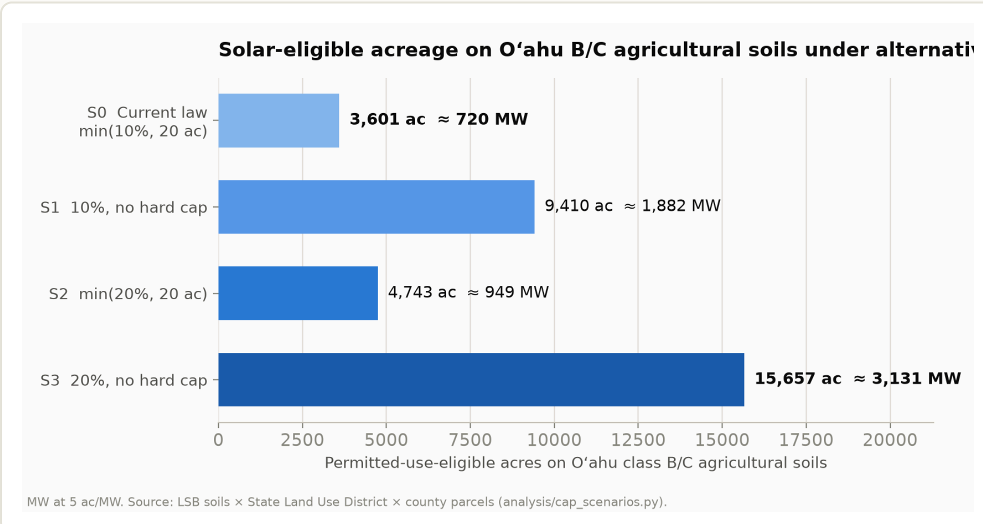

Figure 9. By-right eligible acreage on Oʻahu B/C soils under the current cap and three alternatives. Statewide, the same scenarios run 33,978 → 157,213 acres.

7 Agrivoltaics

Chapter 205 accommodates solar-plus-agriculture at every tier. Within the by-right cap, solar and farming can share a parcel with no further approvals. On the SUP path, ag co-use of the paneled area is a permit *condition* — the statute requires the area be offered for compatible agricultural activities at half of market rent, and grazing, cropping, and apiary uses all qualify. On D/E soils there is no co-use requirement at all. Working examples exist (HARC's agrisolar trial at its Kunia research station has cultivated crops under panels since 2022), and no statutory provision distinguishes agrivoltaic from conventional PV design. The practical constraints on agrivoltaics are economic — racking height, row spacing, and operations coordination — plus the county assessment question in §1: whether co-farmed solar land retains agricultural dedication for the non-paneled area is an administrative practice worth confirming parcel by parcel.

8 Non-agricultural land

Every acre outside the agricultural district falls in one of three other state districts. The rural district is absent from Oʻahu. The conservation district (158,668 acres — the Koʻolau and Waiʻanae ranges) requires a Conservation District Use Permit under HRS ch. 183C for any development; no utility-scale solar has been permitted there, and between watershed subzones, terrain, and permitting practice it is treated here as unavailable. That leaves the **urban district** (104,231 acres), where Chapter 205 imposes no solar restrictions and county zoning governs.

Screening urban parcels for physical candidacy — at least 10 acres at ≤15% slope, non-military, netting out airports, harbors, and golf courses, and keeping parcels whose assessed building value is under 10% of land value (a vacancy proxy from the county assessment roll) — yields **290 parcels totaling 11,319 acres** (13,557 at ≤30% slope), plus roughly 600 net acres of landfills, quarries, and brownfields: about 11,900 acres, or 1,700–2,400 MW at 5–7 acres/MW (Figure 10). The largest concentrations: Kamehameha Schools' Waiawa block (~1,080 flat acres on the 138 kV spine), the Kapolei/Makaiwa Hills area (~740), the Kalaeloa cluster (HCDA, DHHL, Hunt, UH — 900+ non-Navy acres), the Hoʻopili-area D.R. Horton holdings (~780), and Turtle Bay (~450). The special sites are led by the Pacific Aggregate quarry at Wahiawā (150 flat acres), Kapaʻa and Hālawa quarries (89 and 80), Makakilo quarry (80), and Waimānalo Gulch landfill (56). Military installations hold a further ~16,000 flat urban acres, available only through federal enhanced-use leasing.

Physical scarcity of non-agricultural land therefore does not explain solar's concentration on agricultural land: the physical urban envelope rivals the near-grid D/E resource and exceeds current B/C by-right eligibility. The difference is economic and administrative. Median assessed land value in the candidate tier is about $124,000 per acre (75th percentile $1.2M) versus effectively near-zero carrying-cost dedicated ag land, and much of the flat urban inventory is the entitled housing pipeline (Hoʻopili, Makaiwa Hills, Hoakalei) with far higher-value uses in waiting. Classifying every candidate by owner and assessed value (rules in the notes file; pipeline lands are assessed near agricultural rates pre-subdivision, so named-owner identification overrides the value screen) gives three tiers: **durable** — public owners, quarry/landfill/brownfield sites, and private low-improvement land under $50,000/acre — **143 parcels plus 12 special sites, ≈5,660 acres, ~810–1,130 MW**; **uncertain** — mostly Kamehameha Schools, Robinson, and similar holdings at $50k–$500k/acre — 58 parcels, 2,446 acres (~350–490

MW); and **likely higher-value development** — the entitled pipeline (Hoʻopili, Makaiwa Hills, Hoakalei, Hunt/ Kalaeloa, Turtle Bay) — 89 parcels, 3,749 acres. The durable tier is a meaningful complement, not a substitute, for the agricultural district. Wind is categorically unavailable outside AG/Country zoning under Ordinance 25-2 (§9).

Table 9. **Largest durable non-agricultural candidates (full table: data/gis/non**

Site	Owner	Flat acres (≤15%)	Land value $/ac	km to 138 kV	Note
Waiawa	Kamehameha Schools (trust)	598	43k	0.5	Largest du assessed adjacent s
Heʻeia CDD	HCDA	222	19k	0.7	Includes w — field ch
Waimānalo	State of Hawaiʻi	171	—	—	State land
Kapolei mauka	City & County	162	—	—	—
UH West Oʻahu (2 parcels)	University of Hawaiʻi	238	—	—	Existing 1 on adjacer
Pacific Aggregate quarry, Wahiawā	Private (quarry)	150	—	—	Largest da
Kalaeloa + East Kapolei	DHHL	229	—	—	Subject to priorities
Whitmore	HTDC (state)	132	—	—	Ag-tech p
Kalaeloa	State — Dept. of Agriculture	110	—	—	—
Kapaʻa quarry	Private (quarry)	89	—	—	Windward

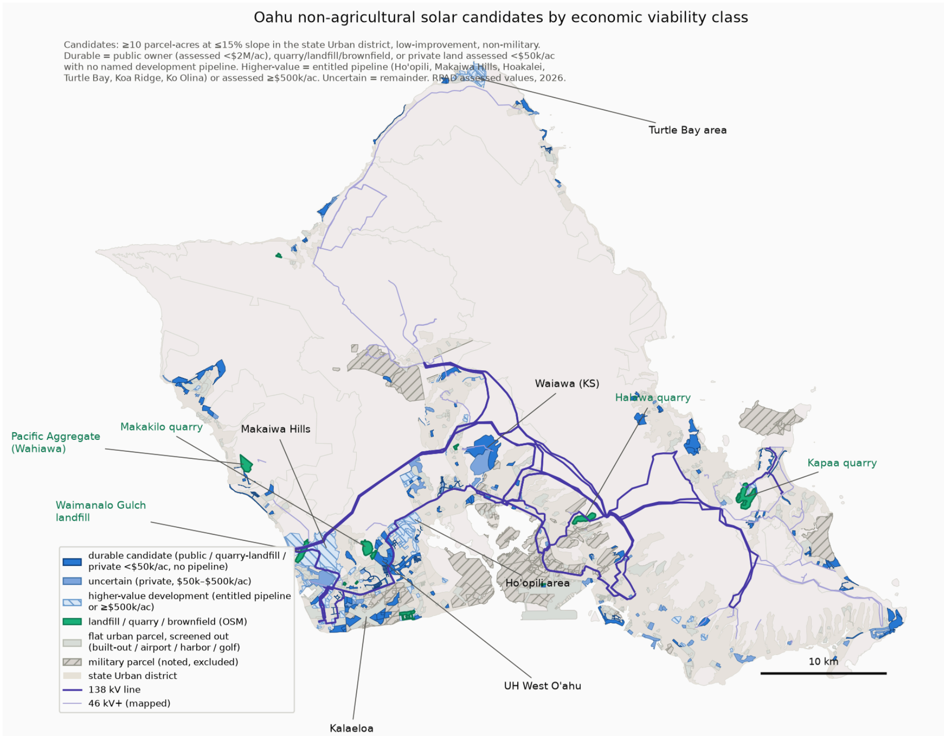

Figure 10. Non-agricultural solar candidates: low-improvement urban-district parcels (≥10 ac at ≤15% slope, non-military), screened-out flat urban parcels, military land (noted, excluded), and landfill/quarry/ brownfield sites, with mapped transmission. Improvement-value proxy and OSM site inventory carry the caveats in the notes file.

9 Wind siting

The state-district rules for wind are permissive: wind energy facilities have been a permitted use on agricultural land of every soil class since 1980, with no acreage cap, subject to a compatibility proviso. The operative constraint is county zoning. Honolulu's Ordinance 25-2 (effective January 2025) sets these standards for turbines of 100 kW and larger:

- Setback of 1× tip height from all property lines, plus the greater of 10× tip height or 1.25 miles from any lot in a country, residential, apartment, apartment mixed-use, or resort district;
- No director's waivers for wind facilities;
- Repowering limited to a 7% height increase (575 ft maximum), and only within the term of the facility's current power-purchase agreement;
- Shadow-flicker (≤10 hr/yr) and noise (≤10 dBA over ambient) limits at pre-existing occupied buildings.

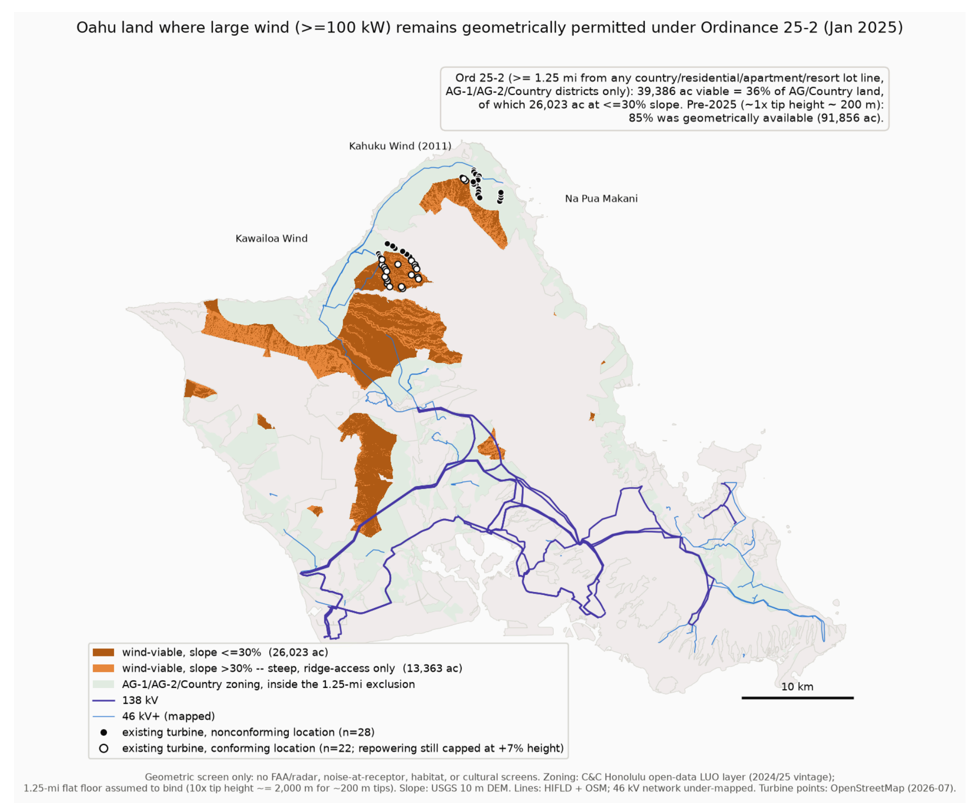

Figure 11. Wind-viable land under Ordinance 25-2: AG/Country zoning beyond 1.25 miles of any country/residential/apartment/resort-district lot, toned by slope, with the three existing (non-repowerable) wind turbines marked individually. Geometric screen only — no FAA/radar, noise-at-receptor, habitat, or cultural screens. **Note on existing turbines inside the viable area:** conformity attaches per turbine, not per farm. Of the 50 existing turbines (OSM locations), 22 stand in the viable area — including 20 of Kawailoa's 30 — while 28 are location-nonconforming: all 8 at Na Pua Makani and 10 of 12 at Kahuku (matching the reported "18 of Kahuku's 20"), plus 10 at Kawailoa (7 inside the setback, 3 on Preservation zoning where large wind is not a permitted use). The expectation that the entire fleet retires does not rest on location: the ordinance permits repowering only within the term of a facility's *current* PPA and caps it at a 7% height increase (575 ft maximum), so every existing machine ages out with its contract regardless of where it stands; conforming-location sites could in principle host newly permitted turbines afterward, subject to the full new-facility standards.

Geometrically, the 1.25-mile rule leaves **39,386 acres — 36% of Oʻahu's 108,417 AG/Country-zoned acres** — eligible, versus 85% under the prior 1×-tip-height standard (Figure 11). Because the Country district is itself protected, every viable acre is AG-zoned, and all of it lies in the mauka north-central interior: the Waialua/Helemano plateau (~21,200 acres, 39% steeper than 30% slope, served only by 46 kV), Kunia (~7,800 acres, mostly gentle, nearest the 138 kV spine), the Kahuku hills (~4,100 acres, majority steep), and Kawailoa/Pūpūkea (~4,000 acres, mostly ≤30% slope). Of the viable total, 26,023 acres are at ≤30% slope. Wind does not displace solar on this land: turbine pads and access roads occupy a few acres per machine — the "small footprint" rationale of the 1980 statute — so viable wind areas overlap the agricultural solar land base, with the main interaction being a modest capacity-factor penalty on co-located PV from tower and rotor shading (directional and calculable; small when arrays sit south of turbines at this latitude). Eighteen of Kahuku's twenty turbines are nonconforming under the new rule; because nonconforming turbines cannot repower beyond their current PPAs, the island's ~155 MW of existing wind is expected to retire as contracts end between 2031 and 2040. For land-screening purposes, large-scale wind on Oʻahu should be treated as foreclosed under current county law; the eligible remainder is concentrated in mauka areas away from residential-district boundaries, where slope and access impose their own limits.

10 State and federal land

Because government owns half the district, public-land disposition is a large share of the siting question. The state's leasing authority is broad: HRS §171-95 (2002) authorizes non-auction leases of up to 65 years to renewable energy producers; §171-95.3 (2009) adds a direct-negotiation process; Act 53 (2024) removed the requirement that output be sold to a utility; and ADC is exempt from public-auction requirements. Use to date, Oʻahu:

- DHHL: the most active state-side lessor (Kalaeloa Solar Two, 2013; subsequent leases at Barbers Point and on Kauaʻi).
- University of Hawaiʻi: 12.5 MW at West Oʻahu (AES, 2024).
- DLNR: its first Oʻahu solar disposition under §171-95.3 (Eurus, 20 MW at Olai) reached Board of Land and Natural Resources hearings in January 2026.
- ADC: hosts solar on Kauaʻi; its Oʻahu holdings (1,700+ idle acres per State Audit 21-01, in the Wahiawā / Whitmore area) have no solar dispositions on record.
- Federal: the military hosts projects under enhanced-use leases (Kupono Solar, ~42 MW on Navy land at Pearl Harbor; HECO's Schofield plant on Army land). Federal parcels also hold the largest block of flat, near-grid D/E land in the screen (Lualualei).

Frictions on state-land leasing are administrative rather than statutory: revenues on ceded-land parcels carry an OHA share; DHHL dispositions answer to beneficiary priorities; ADC's mission is agricultural; and no agency is tasked with a renewable-siting program. Two current processes bear on this: the legislature commissioned a state-lands renewable study in 2025 (HB 778), and HECO is holding renewable-energy-zone workshops on government lands in July 2026.

11 How the rules got here

A short chronology of the operative provisions, for reference; the repository notes contain the full legislative record, including testimony and committee reports for each act.

Table 10. Sources of the current rules

Year	Instrument	Provision
1976	Act 199 (HRS §205-4.5)	Enumerated-use regime on A/B soils; uses not lis 6 special permit
1980	Act 24	Wind added as a permitted use on agricultural lar no cap; language substantively unchanged since
2008	Act 31	Solar added as a permitted use on D/E soils
2011	Act 217 (SB 631)	Solar extended to B/C soils with the min(10% of p class A excluded. The 10% term originated in Ser 20-acre term in House committee (sized to 5 MW for utility competitive-bid waivers)
2014	Acts 52, 55	SUP path above the cap on B/C soils, with the ag decommissioning security, and restoration condit
2015–2025	Seven amendments	Other uses added to the district (hydro, wireless, camps, landfills); solar provisions unchanged sinc
2021	Honolulu assessment practice; Bill 39	Industrial reclassification of solar-farm land; 80% exemption
2025	Honolulu Ordinance 25-2	Current wind setback, waiver, and repowering sta

The testimony record for the solar acts shows the caps were requested by agricultural and planning agencies and the Farm Bureau, with landowners and the solar industry testifying for wider permissions; the wind setback ordinance responded to sustained community opposition at Kahuku. That history is documented in the repository (`notes/sb631-2011.md`, `notes/acts-2014-2022.md`, `notes/wind-setbacks.md`, and related files) and is not developed further here.

12 Data and methods

GIS: Land Study Bureau soil polygons, State Land Use Districts, and county parcels from the state geoportal; ownership from the Honolulu RPAD OWNALL bulk table (99% acreage coverage, entity-resolved); transmission lines from HIFLD and OpenStreetMap, cross-validated (the 46 kV network is under-mapped, so distances are upper bounds); slope from the USGS 3DEP 1/3-arc-second DEM, computed at 10 m in EPSG:26904 with all layers rasterized to a common grid (raster/vector totals reconcile within one acre). Soil-class acreage totals match the Office of Planning's published LSB table to +0.2%. Legal sources: HRS current text and archived versions (statute dating by snapshot diff), Session Laws of Hawaiʻi scans, Honolulu ordinances and council records, complete LUC SP-docket enumeration from files.hawaii.gov. A complete register of exogenous assumptions — including code-level conventions not restated in the text (e.g., minimum cluster-size floors in the corridor and special-site screens, the slope-routing penalty in the expansion heuristic, and the public-land value guard in the viability classification) — is maintained at docs/ASSUMPTIONS.md in the repository, each entry with value, location of use, justification, and disclosure status. Urban vacancy proxy: RPAD assessed building-to-land value ratio (<10%); special sites from OSM (completeness limited). Transmission unit costs: MISO Transmission Cost Estimation Guide, MTEP24 (cdn.misoenergy.org, May 2024) and MTEP25 (June 2025), with an inferred Hawaiʻi construction multiplier flagged UNVERIFIED. Screens not applied: interconnection capacity (confidential), flood, aspect, cultural resources, and county-permit-level constraints. Parcel-level tables (`data/oahu_owner_class_transmission.csv`, `data/oahu_parcel_slope.csv`, `data/sup_census.csv`) are TMK-keyed and analysis-ready.

Known gaps: 46 kV line completeness; PVWatts/SAM aspect runs at 21°N; whether co-farmed solar land retains agricultural dedication for tax purposes; Maui County permit archives (manual pull); the pending SP26-417 decision (~Aug 2026).

Background paper assembled 2026-07-12 by Claude (research agent) for M. Roberts. Every factual claim traces to a primary source cached in the

repository; claims that do not are tagged UNVERIFIED in the notes files and flagged or excluded here. Repository notes: sb631-2011 · wind-statutory-entry · wind-setbacks · pre2011-solar-bills · acts-2014-2022 · sup-census · property-tax-wedge · cap-quantification · oahu-transmission-screen · oahu-ownership · oahu-slope-screen · slope-cost-literature · state-land-solar · synthesis-2026-07-11.